

Muhammad Usman Butt

Software Engineer

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PROFESSIONAL SUMMARY

Software Engineer with experience across game development, AI systems, and full-stack web and mobile. Builds real-time multiplayer and VR experiences in Unity (Netcode, Photon PUN 2, Meta Quest), LLM-driven systems (RAG pipelines, multi-model retrieval, agentic game AI), and production web platforms (React, Node.js, MongoDB, Redis). Placed 5th of 200 teams at the Google Antigravity Mobile Game Challenge 2026 and won 1st place at Softcom 2026. Has led a game development department and shipped cross-platform titles from concept to deployment.

TECHNICAL SKILLS

Languages: C#, Python, JavaScript, TypeScript, HTML, CSS

Game Engine & Multiplayer: Unity (2D/3D), ScriptableObjects, Unity Physics, Animation Systems, State Machines, Object Pooling, Netcode for GameObjects, Photon PUN 2, Unity Relay, Unity Lobby, Host Migration, Reconnection Handling

VR & Immersive Systems: Meta Quest Development, XR Interaction Toolkit, VR Interaction Framework, Final IK, VR Locomotion, Multiplayer VR, Gaussian Splatting

LLMs & RAG: OpenAI API, Groq API, Gemini, NVIDIA NIM, Retrieval-Augmented Generation, Prompt Engineering, Embeddings, Semantic Search, LLM-Driven Game Agents, Procedural Generation

Frontend & Mobile: React, React Native, Android, iOS, WebGL, Cross-Platform Build Pipelines

Backend & Data: Node.js, Express.js, REST APIs, OAuth, BullMQ, MongoDB, Redis, Firebase (Auth, Realtime Database), Cloud Save Systems, Unity IAP, Google AdMob

Tools & Practices: Git, GitHub, CI/CD, Agile/Scrum, Quality Assurance, Team Leadership

PROFESSIONAL EXPERIENCE

AI & Full-Stack Engineer Intern | Termnl Tech

Jun 2026 – Sep 2026

- Built a full-stack RAG platform (React, Node.js/Express, MongoDB, Redis, BullMQ) enabling natural-language queries over connected data sources and uploaded documents, with scoped OAuth integrations and background ingestion jobs.
- Developed features for a React Native Android app, including an AI image generation feature driven by user prompts and in-app selection.
- Worked with the team on quality assurance, testing, debugging, and resolving frontend and feature issues.

Head of Game Development | Axiolink Game Studio

Apr 2026 – Jun 2026

- Led the game development department, directing the full production pipeline from concept to deployment for Unity-based, systems-driven games.
- Set development strategy and coordinated a cross-functional team while remaining hands-on in C# engineering.
- Drove integration of agentic AI into gameplay systems as a core studio direction.

Unity Developer | AstroApe Studios

Sep 2025 – Apr 2026

- Engineered real-time multiplayer features enabling seamless player interactions across game builds.
- Optimized application performance and scalability, ensuring a smooth user experience across platforms.
- Collaborated with a creative team in a hybrid environment, merging artistic vision with technical execution.

Unity Developer | Fiverr (Freelance)

Mar 2023 – Present

- Designed and developed immersive 3D VR experiences using Unity, VR Interaction Framework, Final IK, and Photon PUN 2 for multiplayer environments.
- Solved complex technical challenges to deliver high-quality VR builds aligned with client goals and player expectations.
- Delivered 3+ years of continuous freelance client work across VR and multiplayer game projects.

TECHNICAL PROJECTS

RAG Platform | Full-Stack & AI (Termnl Tech)

Stack: React, Node.js/Express, MongoDB, Redis, BullMQ, OpenAI, Groq

Designed a multi-model retrieval pipeline: OpenAI for answer generation, Groq for low-latency reranking and intent classification. Added Redis caching to cut repeat-query cost and improve response speed.

THRESHOLD | Agentic AI Top-Down Shooter

Stack: Unity, C#, Gemini/Groq/NVIDIA NIM | github.com/usmanbutt-dev/ThresholdGame

Built a 5-agent LLM pipeline driving procedural dungeon generation, adaptive difficulty, and real-time NPC combat decisions (including NPC defection); placed 5th of 200 teams at Google Antigravity Mobile Game Challenge 2026.

ShuffleMania | Mobile Multiplayer Game Collection (Client Project)

Stack: Unity, Netcode for GameObjects, Firebase, Unity IAP, AdMob

Shipped a 5-game mobile collection (3 solitaire variants, real-time multiplayer Ludo and Rummikub) with cloud saves, public/private lobbies, host migration handling, and full monetization (IAP + rewarded ads) across Android and iOS.

A House on Larcombe Road | Co-op Horror, Unity (Client Project)

Stack: Unity, C#, Multiplayer Networking

Built a 2-player psychological co-op horror game with asymmetric twin characters, dual-input puzzle mechanics requiring synchronized actions, and a dynamic sanity system driving vision distortion, stealth detection, and light/sound-based enemy AI.

2D Action Platformer

Stack: Unity, C#

Built a geometry-dash-style 2D action platformer; won 1st place at Softcom 2026 (GIKI).

Unity Modular Inventory System | Open-Source Tool

Stack: Unity, C#, ScriptableObject | github.com/usmanbutt-dev/UnityModularInventorySystem

Designed a reusable, extensible inventory framework using ScriptableObject-based item data and clean data/logic/UI separation for drop-in use across RPG, survival, and crafting games.

For more projects and source code, check out: github.com/usmanbutt-dev

EDUCATION

BS Computer Science | UMT, Lahore | CGPA 3.50/4.00

Aug 2022 – Aug 2026

CERTIFICATIONS

- Game Design and Development 1: 2D Shooter – Michigan State University (Coursera)